Administrator
User Interface
Authentication Controller
User Management Controller
Role Controller
Report Controller
Database
Email Service
Audit Log Service
AUTHENTICATION PROCESS
Enter Login Credentials
Authenticate(username, password)
Query User Table
Return User Record
Verify Password Hash
Check Role Permissions
Return Role: Administrator
Log Login Activity
Authentication Success + Token
Display Admin Dashboard
CREATE NEW USER
Click "Create User"
Request User Creation Form
Get Available Roles
Query Roles Table
Return Roles List
Return Roles
Display User Form
Enter User Details (Login, Email, Role)
Validate User Data
Validate Login Format
Validate Email Format
Check for Duplicate Login/Email
No Duplicate Found
Generate Temporary Password
Hash Password
Insert New User Record
User Created (UserID)
Log User Creation
Insert Audit Record
Send Welcome Email with Credentials
Compose Welcome Message
Email Sent Successfully
Display Success Message
Show "User Created Successfully"
MODIFY USER
Search User
Search Request(criteria)
Query Users Table
Return User List
Display Search Results
Select User to Modify
Get User Details(userID)
Query User Information
Return User Details
Display User Profile
Select "Change Role"
Get Available Roles
Query Roles
Return Roles
Display Role Options
Select New Role & Confirm
Update User Role Request
Check Active Sessions
Return Session Info
Update User Role
Role Updated
Log Role Change
Insert Audit Record (Old Role → New Role)
Notify User of Role Change
Notification Sent
Display Update Success
Show "Role Updated Successfully"
RESET USER PASSWORD
Select "Reset Password"
Request Password Reset
Generate New Random Password
Hash New Password
Update Password & Set ForceChange Flag
Password Updated
Log Password Reset
Insert Audit Record
Send New Password
Compose Password Reset Email
Email Sent
Display Success
Show "Password Reset Successfully"
GENERATE MONTHLY REPORT
Navigate to Reports Section
Request Report Options
Display Report Filters
Select Month & Year
Click "Generate Report"
Generate Monthly Report(month, year)
Query Total Appointments
Return Count
Query Canceled Appointments
Return Count
Query Completed Appointments
Return Count
Query Active Veterinarians
Return Vet Count
Query Schedule Statistics
Return Schedule Data
Calculate Occupancy Rate
Generate Charts & Graphs
Save Report
Report Saved
Return Report Data
Display Report with Visualizations
Click "Export to PDF"
Export Report Request
Generate PDF Document
Return PDF File
Download PDF Report
VIEW AUDIT LOG
Click "Audit Log"
Request Audit Log
Query Audit Table
Return Audit Records
Display Audit Log
Apply Filters (Date, User, Action)
Filter Audit Log(criteria)
Query with Filters
Return Filtered Records
Display Filtered Results
Show Audit Trail
LOGOUT
Click Logout
Logout Request
Log Logout Activity
Invalidate Session Token
Logout Success
Redirect to Login Page
Admiinistrator
User Interface
Authentication Controller
User Management Controller
Role Controller
Report Controller
Database
Email Service
Audit Log Service